3.1 Magmas and Monoids

Definition 3.1.1 (Magma)
*Let X be a set. A **law of composition** on $X \times X$ is a function*

$$\cdot : X \times X \to X$$

and we typically write the composition of $x, y \in X$ as either

$$x \cdot y$$

or xy, or $x + y$ in the commutative case.

*A pair $(X, \cdot)$ consisting of a set X and law of composition on X is called a **magma**.*

Definition 3.1.2 (Magma/Monoid)
*A **magma** $(X, \cdot)$ is said to be*

- ***associative** if $(x \cdot y) \cdot z = x \cdot (y \cdot z)$*
- ***commutative** if $x \cdot y = y \cdot x$ for all $x, y \in X$*
- ***unital** if there exists $e \in X$ such that $e \cdot x = x \cdot e = x$ for all $x \in X$. Such an e is called an **identity**.*
- *a **monoid** if it is both **associative** and **unital***

Proposition 3.1.3 (Identity is Unique)
A magma $(X, \cdot)$ has at most one element e such that

$$x \cdot e = e \cdot x = x$$

for all $x \in X$.

Definition 3.1.4 (Invertible / Monoid)
*Let $(X, \cdot)$ be a unital magma. An element $x \in X$ is **invertible** if there exists $y \in X$ such that*

$$x \cdot y = y \cdot x = e$$

Proposition 3.1.5 (Inverses are unique)
Let $(X, \cdot)$ be a monoid. If $x \in X$ is invertible then its inverse is unique and denoted x^{-1}.

Proof. Suppose that $xy = xy' = e = yx = y'x$. Then

$$xy = e \implies y'(xy) = y'e = y' \implies (y'x)y = y' \implies y = y'$$

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Definition 3.1.6 (Homomorphism)
*Let $(X, \cdot)$, $(Y, \cdot)$ be magmas. Then a function $\varphi : X \to Y$ is said to be a **magma homomorphism** if it satisfies*

$$\varphi(x_1 \cdot x_2) = \varphi(x_1) \cdot \varphi(x_2) \quad \forall x_1, x_2 \in X$$

*If $(X, \cdot)$ and $(Y, \cdot)$ are unital then φ is **unital** if*

$$\varphi(e_X) = e_Y$$

*If $(X, \cdot)$ and $(Y, \cdot)$ are monoids then φ is a **monoid morphism** if it satisfies both these conditions.*

Proposition 3.1.7 ($\mathbb{N}$ is an initial object)
Let $(X, \cdot)$ be a monoid and $x \in X$. Then there is a unique monoid morphism

$$x^{(-)} : (\mathbb{N}, +) \to (X, \cdot)$$

such that

$$x^1 = x$$

Furthermore if $\varphi : (X, \cdot) \to (Y, \cdot)$ is a monoid morphism then

$$\varphi(x^n) = \varphi(x)^n$$

for all $x \in X$ and $x \in \mathbb{N}$.

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Proposition 3.1.8 ($\mathbb{Z}$ is an initial object)
Let $(X, \cdot)$ be a monoid and $x \in X$ be an invertible element. Then there is a unique monoid morphism

$$x^{(-)} : (\mathbb{Z}, +) \to (X, \cdot)$$

such that

$$x^1 = x$$

and

$$x^{-n} \text{ is the inverse of } x^n$$

Furthermore if $\varphi : (X, \cdot) \to (Y, \cdot)$ is monoid morphism then

$$\varphi(x^n) = \varphi(x)^n$$

for all $x \in X$ invertible and $n \in \mathbb{Z}$.

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